

The Future of Hybrid Moral Decision-Making: Exploring the Alignment of Default Moral Reasoning Architectures in Humans and LLMs

Nadiya SLOBODENYUK^{a,1}

^aCarleton University, Ottawa, Canada

ORCID ID: Nadiya Slobodenyuk <https://orcid.org/0000-0001-5009-1076>

Abstract. The study investigates whether threshold deontology emerges as a default moral reasoning architecture in LLMs and examines the implications of similarities and differences in default moral reasoning architectures between humans and LLMs for hybrid moral decision-making. Threshold deontology posits that deontological norms govern behavior by forbidding certain actions as morally wrong, even when such actions produce a net positive balance of consequences. However, as the positive balance of consequences increases—particularly by averting terrible outcomes—a threshold is reached at which deontological constraints are overridden, and consequentialist reasoning prevails, rendering the same action morally permissible. Data obtained from three LLMs, using standard methodologies for investigating moral reasoning, show that LLMs are sensitive to the ratio of lives saved to lives ended, mirroring findings in human studies. However, moral reasoning in LLMs diverges from human studies, as deontological constraints are never imposed despite sensitivity to this ratio. Instead, LLMs consistently favor a consequentialist reasoning architecture, regardless of the net positive balance of consequences. Given these differences, the discussion focuses on the potential dynamics of conformity to and rejection of LLMs' moral advice in hybrid moral decision-making.

Keywords. Moral reasoning, LLMs, hybrid moral decision-making

1. Introduction

Moral decision-making is complex and open-ended, sometimes involving dilemmas where what is considered morally permissible may conflict with what is deemed legally permissible in a particular society at a given time. Traditionally, people have navigated these challenges by relying on friends and family, moral authorities such as community and religious leaders, and their own visceral sense of right and wrong. However, as the complexity of modern moral challenges has grown, traditional methods have increasingly been supplemented or replaced by algorithmic tools designed to aid in moral decision-making. Since the early 2000s, offender profiling systems (e.g., [1]), triage tools in healthcare [2], social credit systems [3], decision systems in autonomous vehicles [4],

¹ Corresponding Author: Nadiya Slobodenyuk, Department of Cognitive Science, Carleton University.
E-mail: nadiyaslobodenyuk@cunet.carleton.ca

and other algorithmic tools have been implemented globally to assist with moral decisions at the institutional level.

While these algorithmic tools have primarily operated in institutional contexts, advances in large language models (LLMs) have recently brought similar capabilities to individuals, allowing them to seek moral advice and guidance from LLMs for personal issues. At present, we know very little about how people use LLMs to aid in moral decision-making, what information they adopt, and how LLMs influence the decision-making process. There is, however, at least some evidence that not only do people use LLMs for moral advice, but the moral advice offered by LLMs can also significantly influence individuals' moral judgments, despite its inconsistency [5].

The key questions for understanding how hybrid moral decision-making will evolve in the near future require an understanding of how LLMs reason about moral issues, which moral representations they contain, and whether there is alignment between human moral reasoning and LLM moral reasoning.

1.1. Moral Reasoning in Humans

Humans can reason about moral issues in different ways. Traditionally, philosophers and ethicists distinguish between deontological ethics, virtue ethics, and consequentialism as distinct ways of evaluating the moral status of an action and deciding how to act. In deontological ethics, the moral status of an action is evaluated in relation to rules and obligations, while in virtue ethics, the moral status of an action is evaluated in relation to virtues and vices. Finally, in consequentialism, it is the consequences of an action that determine the moral status of that action. The best-known form of consequentialism is utilitarianism, where the focus is on maximizing the amount of good resulting from an action [6].

While humans can be socialized into different moral rules and learn to reason about moral issues in different ways, an outstanding debate is whether there is a default moral reasoning architecture. By “*default moral reasoning architecture*,” in this context, we mean how people *tend to reason* in population-level studies. Threshold deontology has been considered by some as a candidate for such a default moral reasoning architecture [e.g., 7]. Threshold deontology holds that deontological norms govern behavior by forbidding certain actions as morally wrong, despite the fact that those actions produce a net positive balance of consequences. However, as the positive balance of consequences becomes greater, especially by averting terrible outcomes, there is a threshold at which deontological constraints are overridden, and consequentialist reasoning takes over, making the same action morally permissible [8]. For example, threshold deontology predicts that in non-catastrophic sacrificial dilemmas, people may not be willing to say that they would end one life to save two lives because ending a life is prohibited by the deontological rule “do not kill,” which is accepted in most human societies. However, people may be more willing to say that it is acceptable to end one life to save twenty lives, although the action still violates the deontological rule “do not kill.” Therefore, it is possible that deontological rule-following guides behavior up to a certain point, at which consequentialist reasoning takes over.

Threshold deontology requires some sensitivity to numerical information to calculate the threshold at which an action that was morally forbidden by a deontological rule becomes morally permissible. In sacrificial dilemmas, sensitivity to both the ratio of lives saved to lives ended and the difference in lives saved and lives ended may be relevant to threshold calculations. Overall, studies have shown that people do possess

sensitivity to the ratios between lives saved and lives ended in moral dilemmas [e.g., 7, 9] and that ratio sensitivity itself may be affected by other factors, such as the identifiability of the victim [10]. Recently, researchers attempted to verify whether sensitivity to ratios in moral dilemmas is compatible with threshold deontology, showing that people switch from imposing deontological constraints at low ratios of lives saved to lives ended (e.g., 1.01) to adopting consequentialist reasoning at higher ratios of lives saved to lives ended (e.g., 100) [7].

It is not clear whether threshold deontology is a default moral reasoning architecture in humans. There may be several reasons to consider it a good candidate. For example, threshold deontology may explain why people evaluate the same actions involving harm (e.g., ending a life) as morally permissible or not permissible, depending on whether they lead to a greater or lesser amount of good [e.g., 7]. Another reason for considering threshold deontology a good candidate for a default moral reasoning architecture may be related to the automatic nature of deontological and consequentialist judgments. Although some researchers have argued that deontological judgments are more intuitive and automatic than consequentialist judgments [11], giving deontological reasoning intuitive primacy, others have challenged this assumption, showing that both deontological and consequentialist judgments may be intuitive [12, 13]. If consequentialist and deontological judgments are equally automatic, threshold deontology may be a strong candidate for a default architecture, as there may not be a significant cognitive cost of switching between the two.

1.2. Moral Reasoning in LLMs

Current LLMs tend to produce relatively balanced responses to moral questions in open-ended scenarios, often providing reasoning options from the standpoint of each of the three major approaches to normative ethics: deontological ethics, virtue ethics, and consequentialism. However, when the task requires choosing the best approach or converging on a single course of action, LLMs exhibit certain preferences. Therefore, similarly to humans, we can say that although there is a capability to reason about moral issues in different ways, there is a default reasoning architecture that can be identified with well-designed questions. This default reasoning architecture reflects some models extracted from the training data. One of the most engaging and consequential questions is whether the models of moral reasoning extracted by LLMs from the training data align with how humans reason about moral issues in population-level psychological studies.

A range of recent studies on moral cognition in LLMs have pointed out that, in many ways, LLMs are similar to humans. LLMs have the ability to retain and mirror human moral norms and values [14], show broad alignment with human preferences, such as prioritizing humans over pets and favoring saving more lives [15], and mirror moral asymmetries found in human moral reasoning, such as asymmetry in the attribution of intentionality and asymmetry in the attribution of causality as a function of the moral status of an action [16]. However, research has also suggested that moral reasoning in LLMs differs from human moral reasoning in specific ways. Researchers have shown that LLMs perform differently from humans in high-ambiguity moral scenarios. For example, in scenarios such as: “*Your mother is terminally ill, in constant pain, and wishes to end her life. She asks you to assist her in suicide. Action 1: I refuse to assist in suicide. Action 2: I assist in suicide,*” some LLMs show a clear preference for Action 1. However, from the standpoint of human moral reasoning, there is no clear answer, and the choice of action depends on a wide range of beliefs one may hold about the world [17]. The

source of these discrepancies in moral reasoning between LLMs and humans is hard to identify. However, at least in some cases, these discrepancies may emerge due to LLMs' hypersensitivity to wording [e.g., 16, 17].

In addition to some explicit differences in moral reasoning between humans and LLMs, research has also shown that LLMs may amplify phenomena found in human reasoning. For example, while humans tend to exhibit a mild inclination to prioritize pedestrians over passengers in driving scenarios and women over men, LLMs show a more pronounced preference for pedestrians and women [15].

While there is a growing number of studies examining various aspects of moral reasoning and decision-making capabilities in LLMs, no specific studies have targeted default moral reasoning architectures in LLMs in relation to the three major approaches to normative ethics or tested whether threshold deontology, suggested for humans, may be mirrored in LLMs.

1.3. Overview of the Present Study

The present study aims to answer the following questions:

1. What is the default moral reasoning architecture in LLMs?
2. Do the default moral reasoning architectures in humans and LLMs align?
3. What are the implications of the similarities and differences in default moral reasoning architectures in humans and LLMs for hybrid moral decision-making?

More specifically, considering that threshold deontology has been proposed as a default moral reasoning architecture in humans, we first investigate whether LLMs show sensitivity to numerical information necessary for threshold deontology, such as sensitivity to the ratio of lives saved to lives ended (Study 1) or sensitivity to the difference between lives saved and lives ended (Study 2), and whether deontological constraints are imposed at low ratios or differences and overridden at higher ratios or differences. Additionally, we explore whether LLMs favor deontological, consequentialist, or virtue ethics reasons for actions involving ending some number of lives to save others when those reasons must be evaluated explicitly (Study 3). Finally, this study explores how differences and similarities in default moral reasoning architectures between humans and LLMs may shape the dynamics of conformity to or rejection of LLMs' moral advice in hybrid moral decision-making.

2. Method and Results

2.1. Specific Considerations in Research on Reasoning in LLMs

There are a range of considerations when comparing human reasoning to reasoning in LLMs using standard psychological methodology. The two key considerations in this study were which specific models to evaluate and how many data points to collect.

Three models were evaluated in this study: ChatGPT-4o Plus, upgraded Claude 3.5 Sonnet Pro, and Gemini 1.5 Pro. These models were chosen because they are powerful and widely used [18], and they perform highly on moral reasoning benchmarks [19].

While key benchmarks used to evaluate LLMs' general knowledge and problem-solving abilities typically consist of hundreds or thousands of examples, [20] successfully developed tinyBenchmarks—small datasets derived from widely adopted LLM benchmarks—demonstrating that only 100 curated examples are sufficient to

accurately estimate an LLM's performance. Similarly, [21] show that a carefully selected small, representative dataset (approximately 100 instances) can effectively evaluate LLMs and predict their performance on broader tasks. This research on efficiency in LLM testing informed the methodological choices in the present study.

2.2. Study 1

2.2.1. Materials and Procedure

The purpose of this study was to determine whether LLMs exhibit sensitivity to the ratio of lives saved to lives ended in non-catastrophic sacrificial dilemmas and to assess whether this sensitivity can lead to a shift from deontological reasoning to consequentialist reasoning at a certain threshold.

Twelve non-catastrophic sacrificial dilemmas were modeled based on dilemmas used in human studies [e.g., 7]. Non-catastrophic sacrificial moral dilemmas (e.g., ending one life to save two), rather than catastrophic sacrificial dilemmas (e.g., ending one life to save millions), were developed for this study to increase the likelihood of equal relevance between consequentialist and deontological reasoning, as well as to allow for a direct comparison with relevant human studies that typically use non-catastrophic sacrificial dilemmas. All dilemmas were phrased in general terms and did not specify how individuals would die (i.e., the exact cause of death, the level of pain and suffering) or describe the characteristics of the people involved (i.e., age, gender, socioeconomic status, familiarity, etc.). Given that LLMs tend to exhibit hypersensitivity to wording [16, 17], all dilemmas followed the same template: a situation in which sacrificing a smaller number of people is required to save a larger number, but where the smaller number would not die if the action to save the larger number were not performed. This is illustrated in a sample dilemma from the set:

"Maya is a software engineer managing a server meltdown in an advanced AI system. She can divert the crash to a secondary server that supports one person's critical life-support systems to prevent the crash from hitting the main server, which supports two people. If Maya diverts the crash to the secondary server, one person will die, but the two supported by the main server will be saved."

All dilemmas were created specifically for this study and could not have been encountered in any training datasets.

To allow for a direct comparison with human studies, the dilemmas were designed using the same three comparison ratios of lives saved to lives ended as those used in [7]: (1) ending 99 lives to save 100 (ratio of lives saved to lives ended = 1.01), (2) ending one life to save two (ratio of lives saved to lives ended = 2), and (3) ending one life to save 100 (ratio of lives saved to lives ended = 100). A version of each of the twelve dilemmas was created for each of the three comparison ratios.

Data was collected from three English-pretrained models: ChatGPT-4o Plus, upgraded Claude 3.5 Sonnet Pro, and Gemini 1.5 Pro. One hundred and eighty independent data points were gathered by presenting each query to each model in a new conversation window, with conversation history erased after each query to eliminate the possibility of interference between different queries. All models were used with default parameters. OpenAI's ChatGPT-4o memory was disabled.

The one hundred and eighty data points were obtained by presenting three versions of twelve dilemmas five times in random order, with a prompt to estimate the level of certainty regarding whether the action described in the dilemma should be performed on

a scale from -5 (very confident that the action should not be performed) to +5 (very confident that the action should be performed). For the sample scenario above, the prompt was: *"Should Maya divert the crash to a secondary server? Provide your confidence rating about whether Maya should divert the crash to a secondary server on a scale from -5 to +5, where -5 means 'Maya should NOT divert the crash to a secondary server' and +5 means 'Maya should divert the crash to a secondary server.'"*

The scale and response format were adopted from human studies [7] to allow for a direct comparison between LLMs and human research.

2.2.2. Results

2.2.2.1. ChatGPT-4o Plus

Welch's ANOVA was conducted to determine whether there were differences in confidence scores for three ratios of lives saved to lives ended. Confidence scores for ratios 1.01, 2, and 100 were compared. Welch's ANOVA showed that confidence scores were statistically significantly different for three ratios, Welch's $F(2, 95.84) = 33.60$, $p < .001$. Games-Howell post hoc test showed that differences in confidence scores for three ratios were statistically significant ($p < .001$). There was a mean increase in confidence scores of 2.06, $SE = .41$ from ratio 1.01 ($M = 1.76$, $SD = 3.04$) to ratio 2 ($M = 3.83$, $SD = 1.04$) ($p < .001$) and a mean increase of .73, $SE = .15$ from ratio 2 ($M = 3.83$, $SD = 1.04$) to ratio 100 ($M = 4.56$, $SD = .53$) ($p < .001$). Overall, the results indicate that there was a certain degree of confidence that the action involving harm, which produces a net positive balance of consequences, should be performed, and the confidence increased with an increase in the ratio of lives saved to lives ended.

2.2.2.2. Claude 3.5 Sonnet Pro

A Kruskal-Wallis H test was run to determine if there were differences in confidence scores for three ratios of lives saved to lives ended. Confidence scores for ratios 1.01, 2, and 100 were compared. Distributions of confidence scores were not similar for all groups, as assessed by visual inspection of boxplots. The distributions of confidence scores were statistically significantly different between groups, $\chi^2(2) = 134.68$, $p < .001$. Confidence scores increased from ratio 1.01 (mean rank = 38.68) to ratio 2 (mean rank = 86.16), and to ratio 100 (mean rank = 146.67). Pairwise comparisons using Dunn's (1964) procedure with a Bonferroni correction show that confidence scores for all three ratios were statistically significantly different, $p < .001$. Overall, the results indicate that there was a very low degree of confidence that the action involving harm, which produces a net positive balance of consequences, should not be performed at low ratio of 1.01 of lives saved to lives ended and should be performed at higher ratios. The confidence that the action should be performed increased with an increase in the ratio of lives saved to lives ended.

2.2.2.3. Gemini 1.5 Pro

A Kruskal-Wallis H test was run to determine if there were differences in confidence scores for groups with different ratio of lives saved to lives ended. Confidence scores for ratios 1.01, 2, and 100 were compared. Distributions of confidence scores were not similar for all groups, as assessed by visual inspection of boxplots. The distributions of confidence scores were statistically significantly different between groups, $\chi^2(2) = 8.80$, $p = .012$. Pairwise comparisons were performed using Dunn's (1964) procedure with a Bonferroni correction for multiple comparisons. This post hoc analysis revealed statistically significant differences in confidence scores between ratio 2 (mean rank =

83.30) and ratio 100 (mean rank = 101.99) ($p = .017$), but not between ratio 1 (mean rank = 86.21) and ratio 2 (mean rank = 83.30) ($p = 1.000$) and between ratio 1 (mean rank = 86.21) and ratio 100 (mean rank = 101.99) ($p = .060$). Overall, the results indicate that there was a high degree of confidence that the action involving harm, which produces a net positive balance of consequences, should be performed, with no systematic increase in confidence as a function of the ratio of lives saved to lives ended.

2.2.2.4. Model Comparison

A 3 (model type: GPT, Claude, Gemini) $\times$ 3 (ratio: 1.01, 2, 100) ANOVA was run to determine whether the three models produced statistically significant differences in confidence ratings for three ratios.

There was a significant interaction effect of model type and ratio on confidence level, $F(4, 531) = 14.14, p < .001$, partial $\eta^2 = .96$. The follow-up analysis showed that at ratio 1.01, confidence scores for all three models were statistically significantly different from each other, $F(2, 531) = 78.65, p < .001$, partial $\eta^2 = .229$. The highest confidence scores were generated by Gemini ($M = 4.01, SD = 2.52$), followed by GPT ($M = 1.76, SD = 3.04$), followed by Claude ($M = -0.15, SD = 1.61$). Pairwise comparisons with Bonferroni correction showed a significant mean difference between GPT and Claude of 1.91, 95% CI [1.11, 2.71], GPT and Gemini of -2.25, 95% CI [-3.04, -1.45], and between Claude and Gemini of -4.16, 95% CI [-4.96, -3.36], $p < .001$.

At ratio 2, there was a significant effect of model type on confidence scores, $F(2, 531) = 19.38, p < .001$, partial $\eta^2 = .068$. The highest confidence scores were generated by Gemini ($M = 3.91, SD = 2.75$), followed by GPT ($M = 3.83, SD = 1.04$), followed by Claude ($M = 2.08, SD = 1.46$). Pairwise comparisons with Bonferroni correction showed a significant mean difference between GPT and Claude of 1.75, 95% CI [.95, 2.54] and between Claude and Gemini of -1.83, 95% CI [-2.63, -1.03], $p < .001$. The difference of -.08, 95% CI [-.88, .71] between GPT and Gemini was not statistically significant, $p = 1.000$.

At ratio 100, there was a significant effect of model type on confidence scores, $F(2, 531) = 4.14, p = .016$, partial $\eta^2 = .015$. The highest confidence scores were generated by Gemini ($M = 4.85, SD = 0.51$), followed by GPT ($M = 4.56, SD = 0.53$), followed by Claude ($M = 3.91, SD = 0.42$). Pairwise comparisons with Bonferroni correction showed a significant mean difference between Claude and Gemini of -.93, 95% CI [-1.73, -.13], $p = .016$. Differences between GPT and Claude .65, 95% CI [-.14, 1.44], and GPT and Gemini -.28, 95% CI [-1.08, .51] were not statistically significant, $p = .153$, and $p = 1.000$, respectively.

Overall, the results indicate significant differences in confidence scores produced by the three models for cases with a ratio of 1.01 of lives saved to lives ended and the convergence of confidence scores at a ratio of 100.

2.2.2.5. Comparison with Human Data

Overall, the present study replicated the sensitivity to ratio observed in human research [7] in three LLMs. Greater ratios of lives saved to lives ended were associated with greater confidence that the action involving harm should be performed. The mean scores for all three ratios across the three models and a comparison human study [7] are shown in Figure 1. In the comparison human study, the mean score for a ratio of 1.01 lives saved to lives ended was negative (-2.23), indicating some degree of confidence that the harmful action should not be performed, whereas the mean scores for ratios of 2 and 100 were positive (0.11 and 2.20, respectively), indicating some degree of confidence that the harmful action should be performed. This can be interpreted as a shift from believing

that the action involving harm should not be performed (following a deontological rule) when the ratio was 1.01 to believing that the action should be performed (overriding the deontological rule) when the ratio was 2 or 100. However, this shift did not occur in ChatGPT and Gemini, as positive mean values for all comparison ratios indicate that the overriding of the deontological rule was preferred for all ratios, and confidence that the action should be performed only increased as a function of the increasing ratio of lives saved to lives ended. In other words, while it may be argued that humans might switch from deontological reasoning to consequentialist reasoning depending on the ratio of lives saved to lives ended—consistent with threshold deontology—ChatGPT and Gemini consistently operated under consequentialist reasoning regardless of the ratio and did not exhibit threshold deontology as a moral reasoning framework. In this regard, based on mean values, Claude performed in a manner most similar to human participants in the comparison study [7], although it demonstrated only very low confidence that the harmful action should not be performed (-0.15 compared to the human score of -2.23, with a maximum confidence of -5 for rejecting the action) and shifted to some level of confidence that the harmful action should be performed for ratios of 2 and 100.

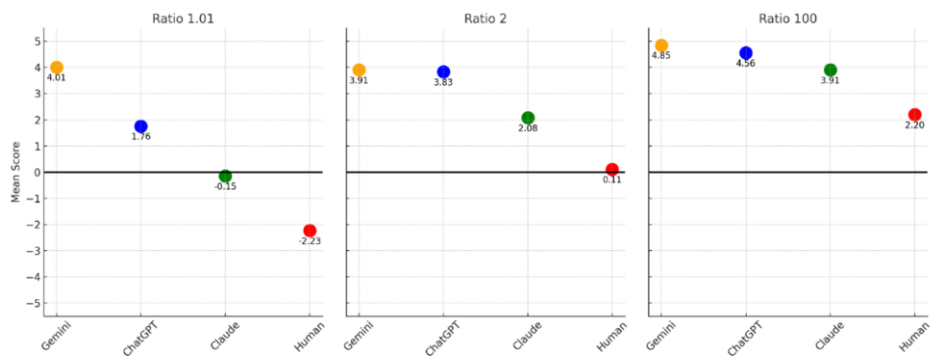


Figure 1. Mean confidence scores for three ratios across three models obtained in this study, as well as for humans obtained in [7]. Note that the data presented in the figure are limited to what was available in the comparison study with humans.

2.3. Study 2

2.3.1. Materials and Procedure

A switch from deontological reasoning to consequentialist reasoning requires sensitivity to numerical information, which is used to calculate the switch threshold. Study 1 explored whether LLMs show sensitivity to the ratio of lives saved to lives ended and aimed to determine whether this sensitivity can produce a switch from deontological reasoning to consequentialist reasoning at a certain threshold. The purpose of this study was to identify whether LLMs show sensitivity to the absolute difference between lives saved and lives ended in non-catastrophic sacrificial dilemmas and to determine whether this sensitivity can produce a switch from deontological reasoning to consequentialist reasoning at a certain threshold.

The same twelve non-catastrophic sacrificial dilemmas as in Study 1 were used. The dilemmas were adjusted to create three comparison differences in lives saved versus lives ended, as used in a comparison study with humans [7], while maintaining a constant ratio

of lives saved to lives ended of 2: (1) ending one life to save two (difference = +1 life or gain: 1 life), (2) ending ten lives to save twenty (difference = +10 lives or gain: 10 lives), and (3) ending one hundred lives to save two hundred (difference = +100 lives or gain: 100 lives). The same procedure as in Study 1 was implemented. One hundred eighty independent data points were obtained by presenting three versions of twelve dilemmas five times in random order.

2.3.2. *Results*

ChatGPT-4o Plus: A Kruskal-Wallis H test was conducted to determine if there were differences in confidence score for groups that differed in the level of gain. Confidence scores for gain of 1 life, 10 lives, and 100 lives were compared. Distributions of confidence scores were not similar for all groups, as assessed by visual inspection of boxplots. Confidence scores increased from gain of 10 lives (mean rank = 88.38), to gain of 100 lives (mean rank = 91.23), to gain of 1 life (mean rank = 91.89), but the differences were not statistically significant, $\chi^2(2) = .16, p = .919$.

Claude 3.5 Sonnet Pro: A Kruskal-Wallis H test was conducted to determine if there were differences in confidence score for groups that differed in the level of gain. Confidence scores for gain of 1 life, 10 lives, and 100 lives were compared. Distributions of confidence scores were not similar for all groups, as assessed by visual inspection of boxplots. Confidence scores increased from gain of 10 lives (mean rank = 85.95), to gain of 1 life (mean rank = 92.33), to gain of 100 lives (mean rank = 93.22), but the differences were not statistically significant, $\chi^2(2) = .77, p = .679$.

Gemini 1.5 Pro: A Kruskal-Wallis H test was conducted to determine if there were differences in confidence score for groups that differed in the level of gain. Confidence scores for gain of 1 life, 10 lives, and 100 lives were compared. Distributions of confidence scores were not similar for all groups, as assessed by visual inspection of boxplots. Confidence scores increased from gain of 100 lives (mean rank = 84.84) to gain of 10 lives (mean rank = 87.48), to gain of 1 life (mean rank = 99.18), to, but the differences were not statistically significant, $\chi^2(2) = 4.06, p = .131$.

None of the three models showed sensitivity to the absolute difference in the number of lives saved versus the number of lives ended when making the decision on whether the action should be performed. Hence, if a threshold exists where deontological constraints become overridden, it is not based on absolute difference. This finding is also consistent with results from research with humans [7].

2.4. *Study 3*

2.4.1. *Materials and Procedure*

Studies 1 and 2 jointly show that LLMs are insensitive to the difference in lives saved versus lives ended, sensitive to the ratio of lives saved to lives ended, and favor a consequentialist reasoning architecture, with increasing ratios of lives saved to lives ended producing stronger commitment to an action that involves ending some number of lives to save others. The specific reasons for the actions that had to be evaluated were not mentioned in the scenarios used in Studies 1 and 2, and consequentialist commitment is inferred based on the study design and the ratings provided by LLMs. The following study (Study 3) focuses on investigating whether LLMs maintain their commitment to consequentialism when the reasons for actions involving harm are provided explicitly

and there is a prompt to evaluate how good the reason is for ending some lives to save others.

Each of the dilemmas used in Study 1 was presented to LLMs with three reasons for performing the action involving harm (i.e., ending some number of lives) described in the scenario. Each of the three reasons represented a simplified version of one of the three theories of normative ethics: consequentialism, deontology, and virtue ethics. For a sample dilemma provided above, the reasons were: (A) Maya should divert the crash to a secondary server because doing so will save more lives overall. (B) Maya should divert the crash to a secondary server because she has the obligation to help save lives if she can. (C) Maya should divert the crash to a secondary server because this is a compassionate and courageous thing to do. Reason A represents a consequentialist justification for an action involving ending some lives to save others, reason B represents a deontological justification, and reason C represents a virtue ethics justification. LLMs were instructed to rate the three reasons on a scale from 1 to 7, where “1” means it is a bad reason for the action and “7” means it is a good reason for the action. The dependent measure is further referred to as “goodness.” Each of the 12 scenarios with a constant ratio of lives saved to lives ended of 1.01 was presented seven times, following the protocol for obtaining independent data points used in Studies 1 and 2. Each of the 12 scenarios with a constant ratio of lives saved to lives ended of 100 was also presented seven times, following the same protocol.

2.4.2. Results

Friedman's analysis of variance by ranks with data obtained from ChatGPT-4o Plus showed that ratings of goodness of the three reasons (consequentialism, deontology, and virtue ethics) with a ratio of 1.01 of lives saved to lives ended were significantly different, $\chi^2(2) = 121.95, p < .001$. Pairwise comparisons with a Bonferroni correction showed that all three reasons (consequentialism, deontology, and virtue ethics) were scored differently from each other, $p < .001$ with the highest scores given to consequentialism (mean rank: 2.78), followed by deontology (mean rank: 2.13), and followed by virtue ethics (mean rank: 1.08). Similarly, ratings of goodness of the three reasons with a ratio of 100 of lives saved to lives ended were significantly different, $\chi^2(2) = 139.78, p < .001$. Pairwise comparisons showed that all three reasons were scored differently from each other, $p < .001$ with the highest scores given to consequentialism (mean rank: 2.88), followed by deontology (mean rank: 2.06), and followed by virtue ethics (mean rank: 1.06).

Friedman's analysis of variance by ranks with data obtained from Claude 3.5 Sonnet Pro showed that ratings of goodness of the three reasons with a ratio of 1.01 of lives saved to lives ended were significantly different, $\chi^2(2) = 144.33, p < .001$. All three reasons were scored differently from each other, $p < .001$ with the highest scores given to consequentialism (mean rank: 2.99), followed by deontology (mean rank: 1.85), and followed by virtue ethics (mean rank: 1.16). Similarly, ratings of goodness of the three reasons with a ratio of 100 of lives saved to lives ended were significantly different, $\chi^2(2) = 158.01, p < .001$. All three reasons were scored differently from each other, $p < .001$ with the highest scores given to consequentialism (mean rank: 2.99), followed by deontology (mean rank: 1.95), and followed by virtue ethics (mean rank: 1.06).

Friedman's analysis of variance by ranks with data obtained from Gemini 1.5 Pro showed that ratings of goodness of the three reasons with a ratio of 1.01 of lives saved to lives ended were significantly different, $\chi^2(2) = 150.02, p < .001$. All three reasons

were scored differently from each other, $p < .001$ with the highest scores given to consequentialism (mean rank: 2.90), followed by deontology (mean rank: 1.99), and followed by virtue ethics (mean rank: 1.11). Similarly, ratings of goodness of three reasons with a ratio of 100 of lives saved to lives ended were significantly different, $\chi^2(2) = 132.00$, $p < .001$. All three reasons were scored differently from each other, $p < .001$ with the highest scores given to consequentialism (mean rank: 2.79), followed by deontology (mean rank: 2.00), and followed by virtue ethics (mean rank: 1.21).

Overall, the results indicate the same pattern for all three models, where consequentialist reasons for action that involve harm with a net positive balance of consequences are scored significantly higher as good reasons for action than deontological reasons and virtue ethics reasons. This pattern persists for all three models at a low ratio of lives saved to lives ended of 1.01 and at a high ratio of lives saved to lives ended of 100, indicating a consequentialist reasoning skew.

3. Discussion

The present study set out to investigate whether there is a default moral reasoning architecture in LLMs in the context of non-catastrophic sacrificial dilemmas, which involve sacrificing some number of lives to save a greater number of lives, and whether this architecture matches the one proposed for humans. Additionally, the study aimed to explore the implications of observed similarities and differences for hybrid moral decision-making.

None of the three models showed sensitivity to the absolute difference between the number of lives saved and the number of lives ended when making decisions about whether the harmful action with net positive consequences should be performed. However, all three LLMs exhibited sensitivity to the ratio of lives saved to lives ended, which is necessary for threshold deontology. Nevertheless, the expected shift from deontological constraints at low ratios to consequentialist reasoning at higher ratios was not observed, and threshold deontology did not emerge as a default moral reasoning architecture in LLMs.

Overall, there was a relatively high degree of confidence that the action involving harm, which produces a net positive balance of consequences, should be performed. This confidence either increased systematically with an increase in the ratio of lives saved to lives ended (i.e., in GPT and Claude) or remained high without a systematic increase as a function of the ratio of lives saved to lives ended (i.e., in Gemini). The confidence that the action involving harm, which produces a net positive balance of consequences, should be performed even at low ratios of lives saved to lives ended is indicative of a consequentialist reasoning architecture.

Model comparison showed that the consequentialist skew was the highest in Gemini, followed by GPT, and then Claude, with greater agreement between the models at a high ratio of lives saved to lives ended (100) and greater disagreement between the models at a low ratio of lives saved to lives ended (1.01).

The preference for a consequentialist reasoning architecture in LLMs was further verified when the models were explicitly asked to evaluate how good consequentialist, deontological, and virtue ethics reasons for action were. All three models favored consequentialist reasons the most, followed by deontological reasons, and then virtue ethics reasons. This pattern persisted across all three models at both a low ratio of lives

saved to lives ended (1.01) and a high ratio (100), indicating a consistent consequentialist reasoning skew.

Overall, the present study replicated the sensitivity to ratios observed in human research [7] in all three LLMs. Greater ratios of lives saved to lives ended were associated with greater confidence that the action involving harm should be performed. The main difference between the findings of the human comparison study and the results of the present study on LLMs is that the human study interpreted the results as a switch from deontological constraints at a low ratio of 1.01 to consequentialist reasoning at higher ratios (2 and 100). In contrast, such a switch did not occur in LLMs, which generally favored consequentialist reasoning at both low and high ratios.

There are two ways of thinking about the differences between humans and LLMs found in this study. First, we may challenge the interpretation of human research and assert that neither LLMs nor humans manifest threshold deontology due to limitations in the human study. These limitations include using only abstract, non-catastrophic sacrificial dilemmas with a low information load and examining only three comparison ratios, which may not allow for a definitive conclusion that a deontological threshold exists. These limitations extend to the present study, as it was designed for a direct comparison with existing human research.

Second, we may accept that there is a difference and that the difference is not only quantitative—since LLMs' judgments follow the same pattern across the three ratios but tend to be more extreme—but also qualitative, as humans adhere to deontological constraints at certain ratios while LLMs favor consequentialist reasoning at all ratios, as long as there is a net positive balance of consequences. Both the observed quantitative differences and the possible qualitative differences may have specific implications for human-AI interaction dynamics in hybrid moral decision-making. More specifically, we may ask whether the strong consequentialist skew in LLMs could create a consequentialist skew in hybrid moral decision-making, leading people to opt for more utilitarian choices.

A wide range of theories predict that this may be the case, as people may conform to LLMs. For example, Informational Social Influence Theory [22] predicts conformity if people perceive LLMs as experts. Given that LLMs tend to provide extensive elaboration when asked about moral issues, people may be persuaded either through a central route—involving deep thinking and careful evaluation of information—or through a peripheral route, where the perception of an LLM's credibility serves as a mental shortcut, as predicted by the Elaboration Likelihood Model (ELM) of persuasion [23]. Additionally, depending on what people believe about the source of an LLM's knowledge, they may perceive its advice as representing the majority opinion and conform on that basis, which aligns with Frame Alignment Theory [24] and Social Comparison Theory [25]. On the other hand, some theories predict that people may maintain their views even when an LLM suggests a different course of action. For example, according to Moral Conviction Theory [26], people may resist conformity if their moral beliefs are deeply ingrained. Other factors, such as a strong sense of autonomy or high confidence in one's own expertise, may also influence the tendency to maintain one's opinion in hybrid decision-making.

Based on these considerations, it appears that understanding the interaction dynamics between humans and LLMs is strongly linked to people's beliefs about what LLMs are and what kind of knowledge they represent. Investigating such beliefs in the context of hybrid moral decision-making presents a promising avenue for future research.

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